PROJECT REPORT GUIDELINES

Students are required to prepare and submit a project report based on the assigned dataset and problem statement. The report should clearly explain the objectives, dataset, methodology, implementation, results, and conclusions of the project.

1. Project Title / Topic

Mention the complete title of the project as given in the problem statement.

Example: Employee Churn Prediction Using Classification Techniques

2. Objectives

Clearly state the objectives of the project.

The objectives should explain:

• What problem is being addressed.

• What the project aims to achieve.

• Which analytical or machine learning technique is being used.

• What type of output or insights are expected.

Use 3–5 concise objectives.

3. Description of the Dataset

Provide a brief description of the dataset used in the project.

Include:

• Dataset name

• Source of the dataset

• Number of records/rows

• Number of attributes/columns

• Description of important attributes

• Data types of the attributes

• Target variable, if applicable

4. Summary of Procedures

Explain the major steps followed in completing the project.

The procedure may include:

1. Loading the required libraries.

2. Importing the dataset.

3. Understanding the structure of the dataset.

4. Checking and handling missing values.

5. Identifying and correcting incorrect or inconsistent data.

6. Removing duplicate records, if required.

7. Performing data preprocessing.

8. Performing exploratory data analysis.

9. Creating appropriate visualizations.

10. Selecting relevant features.

11. Applying the appropriate analytical or machine learning technique.

12. Training the model, if applicable.

13. Making predictions or generating clusters.

14. Evaluating the results.

15. Interpreting the findings.

Students should briefly explain what was done at each stage rather than only listing the commands or code.

5. Evaluation and Results

Present the results obtained from the analysis or model.

Depending on the type of project, include appropriate evaluation measures.

For Classification Projects: Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

For Regression Projects: MAE, MSE, RMSE, and R^2 Score.

For Clustering Projects: Cluster analysis, cluster characteristics, and Silhouette Score, if applicable.

Students should also provide a short interpretation of the results.

> *Example: "The classification model achieved an accuracy of 87%. The results indicate that the model can reasonably identify employees who are likely to leave the organization."*

6. Output Screenshots

Include screenshots of the important outputs generated during the project.

Screenshots may include:

• Dataset overview

• Data cleaning/preprocessing

• Important statistical outputs

• Graphs and visualizations

• Model output

• Evaluation metrics

• Confusion matrix

• Prediction results

• Cluster visualization, where applicable

Each screenshot should have a suitable caption.

Examples: Figure 1: Dataset Overview; Figure 2: Distribution of Employee Attrition; Figure 3: Confusion Matrix of the Classification Model.

Screenshots should be clear, readable, and relevant. Avoid including unnecessary screenshots.

7. Code

Include the complete code used to perform the project.

The code should:

• Be properly formatted.

• Follow the sequence of procedures explained in the report.

• Include necessary comments.

• Be complete and executable.

• Use appropriate variable and function names.

Students should ensure that the code produces the results and screenshots presented in the report.

8. Conclusion

Provide a brief conclusion summarizing the outcome of the project.

The conclusion should mention:

• The major findings from the analysis.

• The performance of the model, where applicable.

• Important insights obtained from the dataset.

• How the project addresses the given problem.

The conclusion should be based on the actual results obtained from the project.

Final Report Structure

1. Project Title / Topic
2. Objectives
3. Description of the Dataset
4. Summary of Procedures
5. Evaluation and Results
6. Output Screenshots
7. Code
8. Conclusion